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Replacement for
-

Image processing systems proof of suitability

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. The names of the persons creating, releasing and checking this Group Standard are filed and can be retrieved in the master data of the TEREK system.

1 Scope and purpose

This supplement to GS 98000 regulates the proof of suitability for camera-based testing and monitoring equipment through the uniform application of a statistical method in capability tests of IP-systems as part of the proof of suitability for sampling inspection. It shall be applied to the use of new systems, changes to the product and changes to the production process that shall be documented in the PLP or series test systems that cannot successfully verify the specifications.

IP-systems, which are designed to determine continuous quantities (e.g. lengths, diameter) are classic test equipment and can be evaluated in terms of test equipment capability using the methods of GS 98000 and VDA Volume 5. Therefore, they are not considered further in this supplement to GS 98000.

This supplement to GS 98000 applies in the entire external supplier network of the BMW Group (and its subsidiaries) and replaces the VA 4.4.7-2 in GS 98000 section 5.7.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

AIAG-VDA FMEA Manual	Failure mode and effects analysis; Design FMEA, process FMEA, supplemental FMEA for monitoring & system response
GS 91011	Special characteristics and L-marking; Identification, designation and handling of the special characteristics and L-marking
GS 98000	Statistical capability studies
GS 98003	FMEA; Failure Modes and effects ANALYSIS
IATF 16949	Quality management system requirements for automotive production and relevant service parts organisations
ISO 9001	Quality management systems; Requirements
VDA Volume 2	Quality management in the automotive industry; Securing the quality of supplies; Production Process and Product Approval (PPF)
VDA Volume 5	Quality management in the automotive industry; Measurement and Inspection processes; Capability, planning and management
VDI/VDE/VDMA 2632 BL3.-1	Machine vision/industrial image processing; Acceptance test of classifying machine vision systems; Test of classification performance

3 Abbreviations, terms and definitions

3.1 Abbreviations

BNV	Folded-normal distribution
BS	Brake disk
IP-system	Image processing system
OK	OK (component is in accordance with the specifications)
IPQ	Integrated Parts Quality
MSA	Measuring system analysis
MUP	Sample inspection
Not OK	Not OK (component does not meet the specifications)
PLP	Production control plan
PM-IT system	System to administrate and monitor test equipment
PPF	Production process and product release (VDA Volume 2)
QMT	Quality management parts

3.2 Terms

There are no terms in this document.

3.3 Definitions

There are no definitions in this document.

3.4 Symbols

C_{ck}	Test process capability index IP-system
$C_{ck, \text{Fehlerschlupf}}$	Test process capability index for undetected errors
$C_{ck \text{ grenz}}$	Limit value for test process capability index
$C_{ck \text{ min}}$	Minimum test process capability index
$C_{ck, \text{ min Fehlerschlupf}}$	Minimum test process capability index for undetected errors
$C_{ck, \text{ Pseudofehler}}$	Test process capability index of pseudo error
C_{mk}	Critical machine capability index
C_{pk}	Critical process capability index
p	Percentage below the threshold T
$p_{\text{Fehler 1. Art}}$	Probability of error type 1
$p_{\text{Fehler 2. Art}}$	Probability of error type 2
$p_{i.O.}$	Probability of OK parts
$p_{n.i.O.}$	Probability of not OK parts
$p_{\text{Pseudofehler}}$	Probability of pseudo error
$p_{\text{unerkannter Fehler}}$	Probability of undetected errors
u_{1-p}	Standardized normal distribution
u_{α}	With $\alpha = 1-p$ is the α -Quantile of standardized normal distribution

4 Designation

There is no designation in this document.

5 Process and responsibilities

The responsibilities for preparing the proof of suitability are vested in the supplier. It shall be ensured that the responsible persons can prove their competence for the task involved. The proof of suitability certificates shall be submitted to BMW as part of the PPF process. Methodically deviating suitability certificates shall be justified and confirmed by the specialist department with the support of the responsible QMT. This shall be documented in the IPQ module "MUP - Sample Inspection".

6 Proof of suitability for quality feature monitoring IP-systems

6.1 General

The MSA in classifying image processing refers to the evaluation of reproducibility and reliability of an IP-system for the automatic classification of objects. The aim is to ensure that the system works precisely at all times.

The MSA comprises a number of steps, including data collection and analysis, the examination of systematic and random error and the analysis of the environmental conditions.

Data collection and analysis comprises the measurement of classification accuracy for a range of reference objects. The statistical indicators such as average and standard deviation shall be recorded. The analysis of systematic errors such as distortions due to lighting and resolution shall be carried out.

Random errors shall be analyzed by evaluating the reproducibility of the classification function. Variance analysis methods can be used for this purpose. The analysis of environmental conditions shall include factors such as lighting, background and number of reference objects.

NOTE 1 Performing an MSA is an important step to ensure that the classifying IP-system works effectively, accurately and reliably and therefore meets the quality requirements.

IP-systems that are used to monitor quality features generally fall into the class of testing and monitoring equipment in accordance with ISO 9001 and IATF 16949.

According to this supplement to GS 98000, proof of suitability shall be provided for these IP-systems by means of a capability certificate. Continued suitability shall be provided in accordance with section 6.6.

Risk-based validation with regard to the probability of occurrence of the P-FMEA in accordance with GS 98003 shall be ensured. When validating safety-critical or legally required quality characteristics via IP-systems, an increase in the $C_{ck\ min} = 2,0$ is recommended due to the susceptibility to errors.

NOTE 2 A large proportion of IP-systems are designed as comparative (classifying) systems which compare the observed images with features extracted from target images (by "teaching"). Ultimately, such an IP-system assigns the properties "OK" or "not OK" to the components to be tested, which corresponds to an attributive test. Given the expected or typically required low error rates, it is not appropriate to evaluate the IP-system using the methods for attributive systems such as hypothesis testing/Fleiss kappa, as the number of necessary tests would be correspondingly high. In addition, the attributive approach does not provide any clues as to how close the decision between "OK" and "not OK" was in each case, i.e. how robust the monitoring is realized (see VDI/VDE/VDMA 2632 BL3.1).

The IP-system shall use one or more indicator values for the evaluation, which is determined using an algorithm specific to the application case and compared with one or more threshold values in order to assign a property such as "OK" or "not OK".

The indicator represents a measured value that can be suitable for a statistical investigation. For this purpose, it shall have sufficient resolution so that, in the case of purely random influences, this is normally distributed around its expected value or, in the case of limited indicators, it has a folded normal distribution.

If several error patterns are evaluated and sorted by an IP-system, the IP-system shall be seen as a parallel system and each error pattern shall be evaluated and released separately.

The "golden rule of measuring technology" may not be used for IP-systems. The capability for sorting with the required accuracy shall be verified for the IP-system.

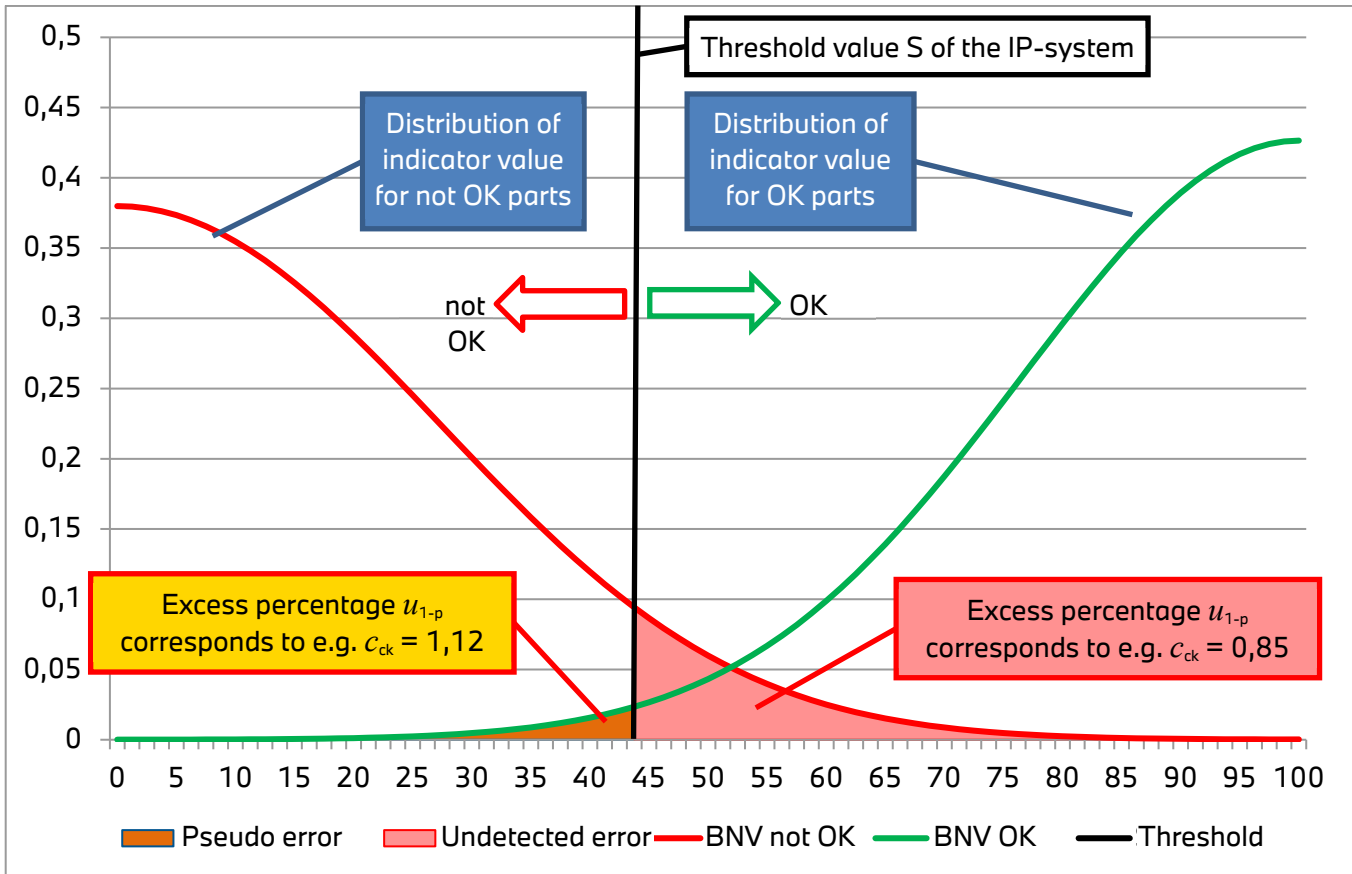
In principle, the following errors are possible in a sorting process with two categories "OK" and "not OK":

Error type 1: An "OK" component is evaluated as "not OK" → pseudo error

Error type 2: A "not OK" component is evaluated as "OK" → undetected error

EXAMPLE Folded normal distribution features (BNV)

NOTE 3 The areas are not shown to scale, but are significantly enlarged for clarification. Such a system would not have sufficient sorting capacity.

**6.2 Undetected load cell capability**

Error type 2 is the more serious error in terms of customer supply conformity.

Error type 2 is calculated as an "and" function and thus multiplication of the individual probabilities with the following equation.

$$p_{\text{unerkannter Fehler}} = p_{\text{n.i.O.}} \times p_{\text{Fehler 2. Art}}$$

Where:

$p_{\text{unerkannter Fehler}}$	probability of undetected errors;
$p_{\text{n.i.O.}}$	probability of not OK parts;
$p_{\text{Fehler 2. Art}}$	probability of error type 2.

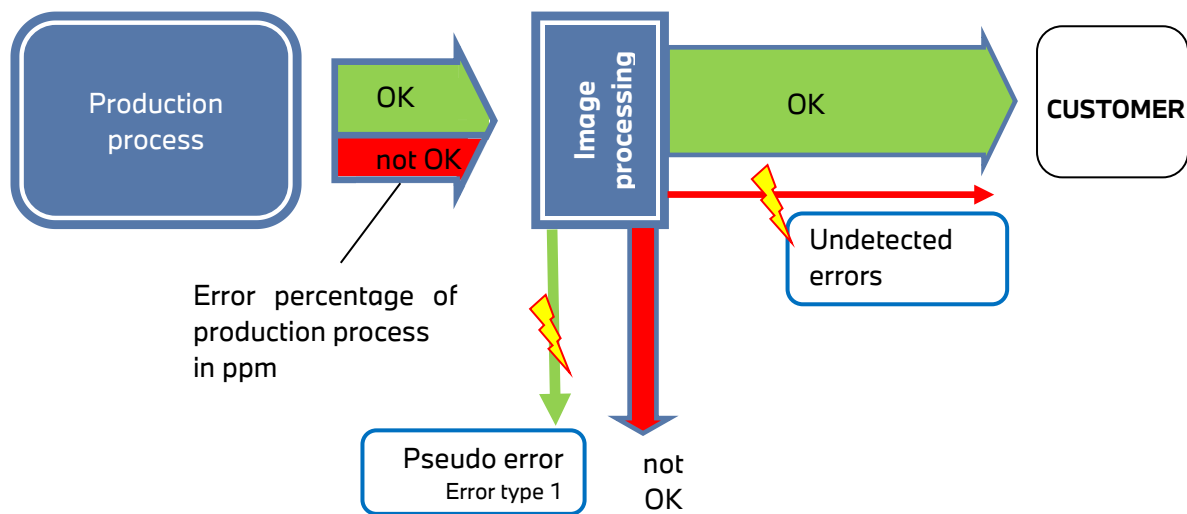


Figure 1: Schematic representation of the interplay between production and testing process

EXAMPLE 1 A production process has a proportion of 500 ppm of faulty components and the IP-system has an undetected error of 2 %, this results in an expected error rate for the undetected errors of

$$500 \text{ ppm} \cdot 0,02 = 10 \text{ ppm}$$

To assess sorting ability, undetected errors are used as indicator and converted into the test capability index c_{ck} of a normally distributed comparison process.

$$c_{ck} = \frac{u_{1-p}}{3}$$

Where:

c_{ck} test process capability index IP-system;
 p percentage below the threshold T;
 u_{α} with $\alpha = 1-p$ is the α -Quantile of standardized normal distribution.

Since not all factors influencing the indicator value will take effect in the course of a short-term analysis similar to machine capability in manufacturing machines, the requirement for the IP-system is designed in such a way that the proportion of faulty parts due to the undetected errors to the customer corresponds to that of a manufacturing process with a manufacturing machine at a $c_{mk} = 1,67$.

The fewer errors are created by the production process, the lower are the requirements posed on the sorting capability.

Depending on the maximum expected percentage of errors in the production process, this results in the following minimum values $c_{ck \text{ min}}$ see Table 1.

If this percentage of errors cannot be estimated on the basis of comparable processes, it is sufficient to classify the probability of occurrence of errors from the associated P-FMEA:

Table 1: Limit value and maximum undetected errors

Occurrence A in the P-FMEA	Error percentage of production process with reference to AIAG-VDA FMEA manual) / prediction of error cause over time		maximum undetected errors of IP-system	Limit value $C_{ck\ min, Fehlerschlupf}$
(with reference to AIAG-VDA FMEA manual)			(see Annex B)	
4	> 100 ppm	(more than 1 x per year)	3 000 ppm	0,92
5	> 250 ppm		1 200 ppm	1,01
5	> 500 ppm	(more than 1 x per month)	600 ppm	1,08
6	> 1 000 ppm		300 ppm	1,14
6	> 2 000 ppm	(more than 1 x per week)	150 ppm	1,21
7	> 10 000 ppm	(more than 1 x per day)	60 ppm	1,28

For error rates that lie between the table values, the next higher one should be selected as the limit value.

NOTE Application in case of $A > 7$ is possible but not practical. If $A > 7$ error avoidance should be improved first, because otherwise the requirements for the IP-system are extremely high.

EXAMPLE 2 $C_{ck\ min}$

With an expected error percentage of 700 ppm in the production process, the limit value $C_{ck\ min, Fehlerschlupf} = 1,14$ corresponds to the table value for 1 000 ppm.

The IP-system is capable with regard to undetected errors if the following condition is met:

$$C_{ck, Fehlerschlupf} \geq C_{ck\ min, Fehlerschlupf} \text{ (see Table 1)}$$

Where:

$C_{ck, Fehlerschlupf}$

test process capability index for undetected errors;

$C_{ck\ min, undetected}$

minimum test process capability index for undetected errors.

As the quality performance of the manufacturing process can often only be estimated in practice and this estimate is subject to uncertainty, particularly in the case of new manufacturing processes, the **"safe launch"** procedure is recommended. This means that the IP-system is used parallel to a **second, reliable 100 % quality control** for a defined period of time (e.g. manual, visual expert control) to be able to register and correct any wrong decisions that occur.

In practice, the **indicator values** will often not have a folded/normal distribution, as there is an overlapping of distributions depending on the algorithm used to calculate the indicator values. It is important that the results obtained in the test are representative, in particular with regard to the least favorable indicator values, i.e. the values closest to the threshold value were recorded in the test (see examples in Annex A).

6.3 Capability for pseudo-errors

In case of the 1. type the situation is slightly different.

Since the probability shall be calculated here under the condition "OK" - component is tested -, the following formula is obtained:

$$p_{\text{Fehlerschlupf}} = p_{\text{OK}} \times p_{\text{Fehler 1. Art}} \quad (6.3.1)$$

Where:

$p_{\text{Pseudofehler}}$	probability of pseudo error;
$p_{\text{i.o.}}$	probability of OK parts;
$p_{\text{Fehler 1. Art}}$	probability of error type 1.

In practice, this means that the pseudo-error rate of the IP-system corresponds almost one to one to the pseudo error rate, which leads to rejections and inspections in the production process ($p_{\text{i.o.}} \approx 1$). This shall therefore often be smaller than the permissible undetected errors in order to limit unnecessary rework costs.

With an acceptable rate of one pseudo error per 1 500 components (= 667 ppm), there is a requirement for a long-term capability value of $c_{\text{ck}} \geq 1,07$.

To ensure this, the following limit value is required for the **short-term test**, similar to the machine capability, since significant influencing factors do not take effect during the short-term investigation:

$$c_{\text{ck, Pseudofehler}} \geq 1,33 \quad (6.2)$$

Where:

$c_{\text{ck, Pseudofehler}}$	test process capability index of pseudo error.
-------------------------------	--

$$c_{\text{ck}} = \frac{u_{1-p}}{3} \quad (6.3)$$

Where:

c_{ck}	test process capability index IP-system;
p	percentage below the threshold T;
u_{α}	with $\alpha = 1-p$ is the α -Quantile of the standardized normal distribution.

6.4 Alternative limit value analysis

As an alternative to the analysis based on the c_{ck} value, the undetected errors and (pr the pseudo error rate can also be taken into account as excess percentage in ppm with the above-mentioned limit values (see sections 6.2 and 6.3).

6.5 Test program to determine the capabilities

The suitability of an attributive test process in which optical sensors and IP-systems are used is evaluated by means of a reference test lot.

It always includes test objects and additional reference evaluations if correctness and not just the repeatability of the test process is to be evaluated. Reference evaluations are either based on a variable counter measurement, or if this is not possible on the reference decision of the reference tester or a reference test team. The appropriate and comprehensive documentation of the reference test lot is mandatory.

The reference test lot should contain both reliable reference as well as limit samples. The suitability of the test process can only be assessed if the reference test lot represents all the characteristics to be evaluated and contains a sufficient number of them.

The composition and size of the reference test lot shall be defined by suitably trained and responsible personnel.

The reference test lot or its test objects may not be used for the development of the test system as otherwise no objective proof of suitability can be provided.

The following test program shall be used to determine the appropriate capability indicators:

Pseudo error:

The suitability shall be determined for at least 50 OK parts. If the data of the indicators is not normally distributed, larger samples can be required.

In the case of image processing tests already used in series production, the ability to identify pseudo errors can also be proven on the basis of appropriate error/rework statistics. In the case of more than 1 pseudo error per 1 500 produced parts or $c_{ck, \text{Pseudofehler}} < 1,33$ the process shall be improved.

Undetected errors:

Use of at least 50 measurements on representative not OK parts or - if not enough are available - also on failure simulation parts (if possible 50 parts or if not as many are available, then, for example, 10 parts of 5 repetitive measurements, unless the repetitions result in identical values).

The use of less than 10 parts is only permitted in justified exceptional cases in consultation with the QMT.

6.6 Proof of continued suitability

According to ISO 9001, the monitoring equipment, which includes the IP-systems, shall be maintained (maintenance and adjustment) in such a way as to ensure continued suitability. A proof shall be provided for this.

IP-systems shall be regularly checked for functionality, whereby the frequency and test method depends on the risk posed by the potential failure of the IP-system (see section "failure safety" in IATF 16949).

A (Annex informative) Calculation of c_{ck} undetected errors / pseudo errors of different use cases

When performing the proof of suitability (measurement system analysis - MSA) for camera systems, consideration must generally be given to the occurrence of undetected errors (type 2 errors) and pseudo errors (type 1 errors).

In view of the fact that established statistical software often does not take into account the reporting of the normal distribution of amounts, alternative methods should be used here.

One practical method is the use of a histogram evaluation including the switching threshold for an initial indication. Afterwards, the c_{ck} value is calculated.

For this purpose, three examples are presented here.

1. Example "conformity of inner diameter" - combined with section 6.2 (undetected errors) and section 6.3 (pseudo errors)

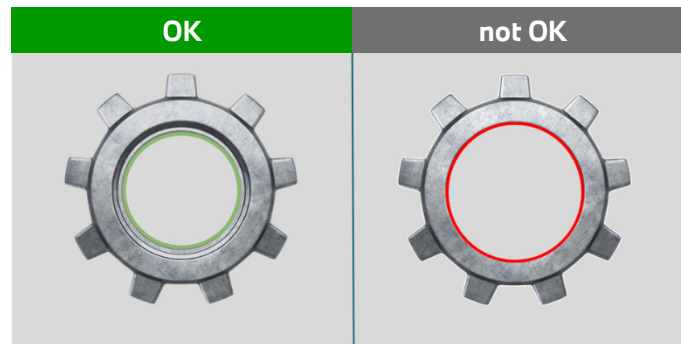
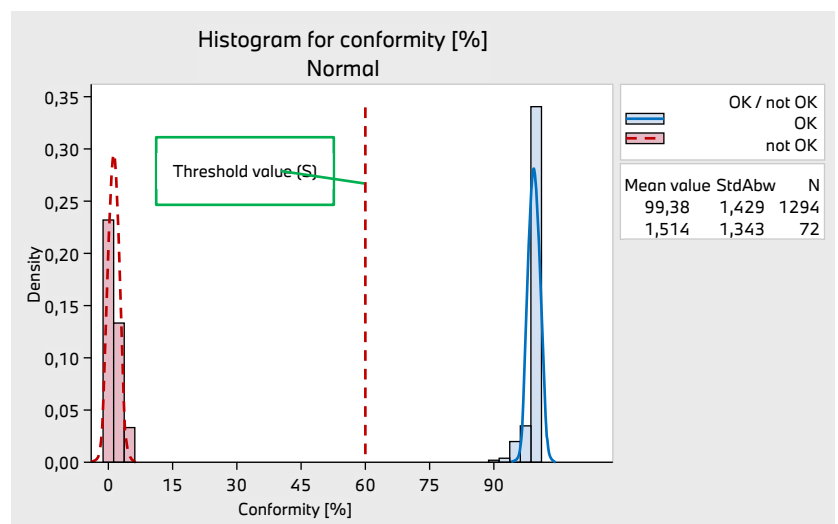


Figure A. 1: Conformity of inner diameter (exemplary)

The validation of a gear was analyzed based on the degree of conformity of the inner diameter in % (indicator). In this case, 100 % was the verification of the small and 0 % the rejection of the gear with the large inner diameter.

Influences for generating uncertainties in the evaluations can be stray lights during the test, changes in position or soiling of the test object. As a result, there are test values < 100 % und > 0 %.

Check	Accordance [%]	OK / not OK
414	100	OK
415	100	OK
416	93	OK
417	100	OK
418	100	OK
419	100	OK
420	100	OK
421	100	OK
422	97	OK
423	100	OK
424	99	OK
425	99	OK
426	98	OK
427	96	OK
428	100	OK
1295	2	not OK
1296	1	not OK
1297	2	not OK
1298	1	not OK
1299	2	not OK
1300	0	not OK
1301	1	not OK
1302	3	not OK
1303	1	not OK
1304	1	not OK
1305	0	not OK



Excerpt of data analysis of suitability verification

A total of 1 366 tests were performed. 1 294 Tests (N) were found to be "OK" with a conformity spectrum in % of 88 to 100. This results in a mean value ($\bar{x}_{\text{across}}$) of 99,38 and a standard deviation (StdAbw) of 1,429 (determined by the statistics program).

The 72 correctly identified "not OK" specimens (N) with a conformity spectrum in % of 0 to 5 resulted in a mean value ($\bar{x}_{\text{across}}$) of 1,514 and a standard deviation (StdAbw) of 1,343.

For example, the switching threshold (S) is not set at 60 % of conformity to ensure sufficient reliability so that outliers only have an insignificant influence on the test result.

Visually, this makes it easy to see that both type 1 errors (pseudo errors) and type 2 errors (undetected errors) can hardly occur with this test system.

For precise verification, the c_{ck} is now calculated on the basis of the available data.

The following formula is used $c_{\text{ck}} = \frac{u_{1-p}}{3}$.

u_{1-p} results from the difference between the threshold (S) and the means value ($\bar{x}_{\text{across}}$) divided by the standard deviations (StdAbw).

Calculation error type 1 (pseudo error):

$$u_{1-p} = (99,38 - 60) / 1,429 = 27,56 \quad c_{\text{ck}} = 27,56 / 3 = 9,19$$

$$c_{\text{ck}} > c_{\text{ck, min}} = \text{passed}$$

Calculation error type 2 (undetected error):

$$u_{1-p} = (60 - 1,514) / 1,343 = 43,55 \quad c_{\text{ck}} = 43,55 / 3 = 14,52$$

$$c_{\text{ck}} > c_{\text{ck, min}} = \text{passed}$$

Error type	Conformity [%] $\bar{x}_{\text{across}}$	Standard deviation (StdAbw)	Threshold (S)	Difference (S to $\bar{x}_{\text{across}}$)	Difference in StdAbw units	Capability index c_{ck}	Requirement $c_{\text{ck, min}}$	Result
Pseudo error	99,38	1,429	60	39,38	27,56	9,19	1,33	passed
Undetected errors	1,514	1,343	60	58,49	43,55	14,52	1,28	passed

P-FMEA evaluation with A = 7 (see Table 1); $c_{\text{ck, min}}$ Fehlerschlupf: 1,28

2. Example "size in pixel" - combined with section 6.2 (undetected error) and section 6.3(pseudo error)

The diameter of an installed brake disc (533 Px) was compared with the next smallest size (511 Px) based on the size (diameter) in pixels (indicator) to avoid mix-ups during installation.

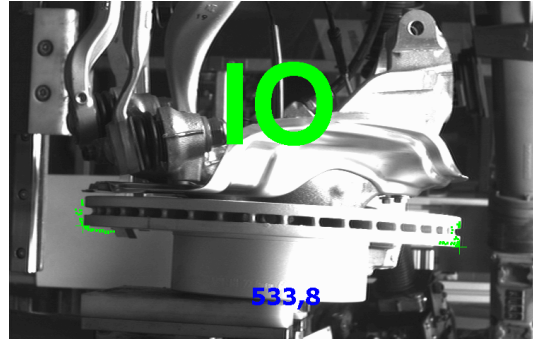
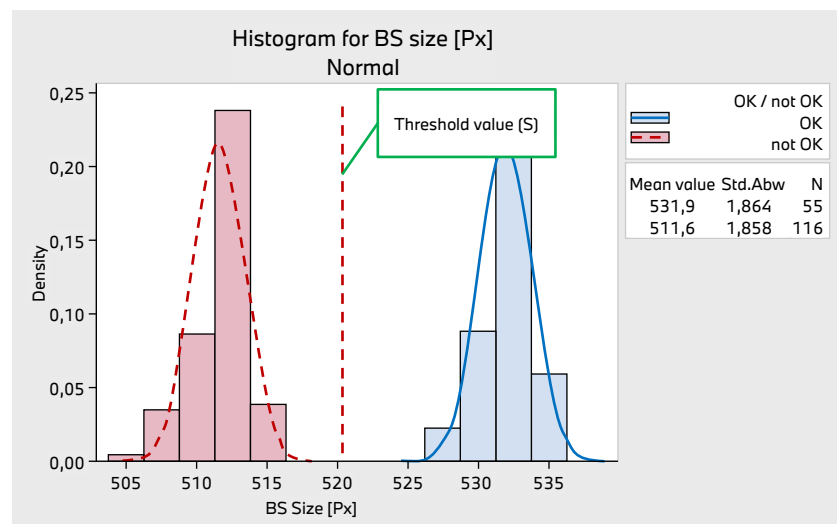


Figure A. 2: Conformity of installed brake disk (exemplary)

Check	BS Size [Px]	OK / not OK
106	512,5	not OK
107	513,6	not OK
108	510,5	not OK
109	512,6	not OK
110	513,4	not OK
111	509,4	not OK
112	511,7	not OK
113	509,6	not OK
114	510,5	not OK
115	508,4	not OK
116	508,4	not OK
117	532,7	OK
118	532,3	OK
119	532,8	OK
120	529,7	OK
121	533,7	OK
122	529,5	OK



Excerpt of data analysis of suitability verification

for this purpose, 171 tests were carried out with the result that 55 were evaluated correctly as "OK" and 116 as "not OK".

The mean value ($\bar{x}$) for the OK tests is 531,9 with a standard deviation (StdAbw) of 1,864. The mean value ($\bar{x}$) for the "not OK" tests is 511,6 and the StdAbw is 1,858.

The threshold value (S) was here determined at 521 Px.

Using the calculation formula $c_{ck} = \frac{u_{1-p}}{3}$ the result is as follows:

Calculation error type 1 (pseudo error):

$$u_{1-p} = (531,9 - 521) / 1,864 = 5,85 \quad c_{ck} = 5,85 / 3 = 1,95$$

$$c_{ck} > c_{ck, \min} = \text{passed}$$

Calculation error type 2 (undetected error):

$$u_{1-p} = (521 - 511,6) / 1,858 = 5,06 \quad c_{ck} = 5,06 / 3 = 1,69$$

$$c_{ck} > c_{ck, \min} = \text{passed}$$

Error type	Size in Px ($\bar{x}$)	Standard deviation (StdAbw)	Threshold (S)	Difference (S to $\bar{x}$)	Difference in StdAbw units	Capability index c_{ck}	Requirement $c_{ck, \min}$	Result
Pseudo-error	531,9	1,864	521	10,9	5,847639485	1,95	1,33	passed
Undetected errors	511,6	1,858	521	9,4	5,059203445	1,69	1,28	passed

P-FMEA evaluation with A = 7 (see Table 1); $c_{ck, \min}$ Fehlerschlupf: 1,28

This example clearly shows that the safety margin to the threshold value is not very large for both error type 1 and error type 2 although the camera system provides good safety for both with c_{ck} 1,95 and 1,69. Nevertheless, care should be taken to ensure that the upstream production process operates with a stable error rate and that process monitoring ensures effective control.

3. Example "size in pixel_pseudo error" - combined with section 6.2 (undetected error) and section 6.3(pseudo error)

The diameter of an installed brake disc (533 Px) was compared with the next smallest size (511 Px) based on the size (diameter) in pixels (indicator) to avoid mix-ups during installation. Due to the influence of stray light, three tests of "OK" parts (533 Px) resulted in incorrect evaluations and rejection by the IP-system, as it only recognized values of ~ 520 Px.

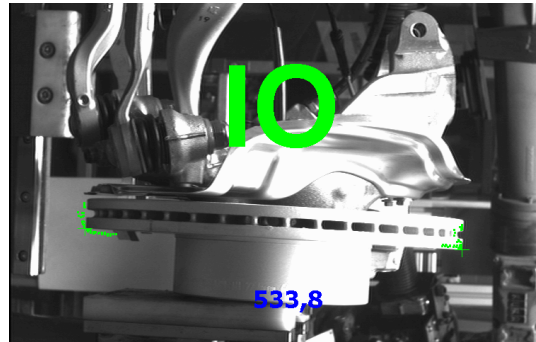
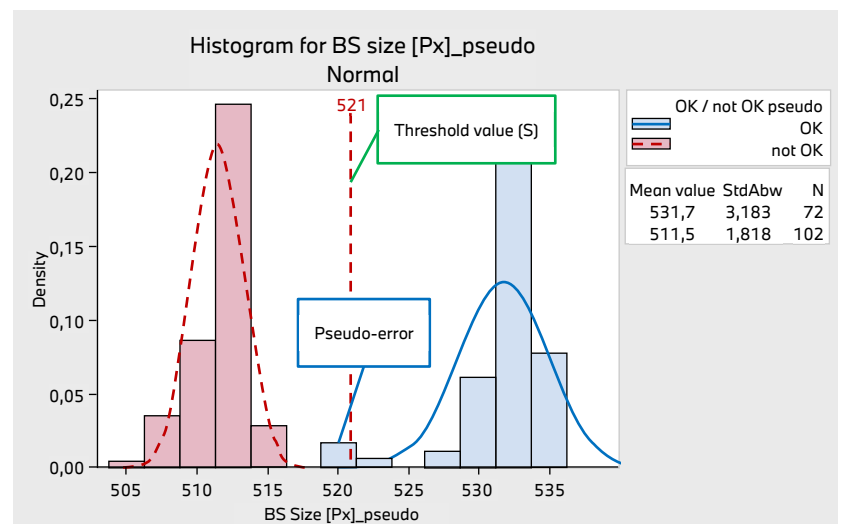


Figure A. 3: Conformity of installed brake disk (exemplary)

Check	BS Size [Px]_pseudo	OK / not OK pseudo
94	510,5	not OK
95	512,6	not OK
96	513,4	not OK
97	509,4	not OK
98	511,7	not OK
99	509,6	not OK
100	510,5	not OK
101	508,4	not OK
102	508,4	not OK
129	531,5	OK
130	533,3	OK
131	533,5	OK
132	531,7	OK
133	534,1	OK
134	520,1	OK
135	520,9	OK
136	532,6	OK
137	533,8	OK
138	527,4	OK
139	534,8	OK



Excerpt of data analysis of suitability verification

For this purpose, 174 tests were carried out with the result that 69 were evaluated correctly as "OK" and 102 as "not OK".

The threshold value (S) was determined at 521 Px.

Components identified as "OK" had a range of (527 to 536) Px. However, there were also 3 test specimens which were rated as "not OK" by the IP-system due to interference with values of 520 Px (pseudo error).

"Not OK" components with a range of (505 to 515) Px were safely removed.

The mean value ($\bar{x}$) for the OK tests is 531,7 with a standard deviation (StdAbw) of 3,183. The mean value ($\bar{x}$) of the "not OK" tests is 511,5 and the StdAbw is 1,818.

Using the calculation formula $c_{ck} = \frac{u_{1-p}}{3}$ the result is as follows.

Calculation error type 1 (pseudo error):

$$u_{1-p} = (531,7 - 521) / 3,183 = 3,36$$

$$c_{ck} = 3,36 / 3 = 1,12$$

$$c_{ck} > c_{ck, \min} = \text{not passed}$$

Calculation error type 2 (undetected error):

$$u_{1-p} = (521 - 511,5) / 1,818 = 5,23$$

$$c_{ck} = 5,23 / 3 = 1,74$$

$$c_{ck} > c_{ck, \min} = \text{passed}$$

Error type	Number of "not OK" Px ($\bar{x}_{across}$)	Standard deviation (StdAbw)	Threshold (S)	Difference (S to $\bar{x}_{across}$)	Difference in StdAbw units	Capability index c_{ck}	Requirement $c_{ck, \min}$	Result
Pseudo-error	531,7	3,183	521	10,70	3,36	1,12	1,33	not passed
Undetected errors	511,5	1,818	521	9,50	5,23	1,74	1,28	passed

P-FMEA evaluation with A = 7 (see Table 1); $c_{ck, \min}$ Fehlerschlupf: 1,28

In the case of pseudo errors, which lead to the result "not passed", versed experts / testers in the quality control circle shall evaluate these errors. Subsequently, the IP-system shall be optimized and the test / release repeated.

B (Annex informative) Calculation of c_{ck} undetected errors in case of different error rates of the production process

With an expected maximum error rate of 500 ppm a c_{ck} limit value of 1,08 was determined. If the error rate of the production process deviates from this, the following calculation can be carried out to obtain the then valid c_{ck} limit value:

With a normal distribution, a c_{ck} of 1,08 corresponds to an excess of 598 ppm (see Tables for normal distribution at $u = 3 \cdot c_{ck \text{ grenz}} = 3 \cdot 1,08 = 3,24$).

In order to achieve the same results with different error rates, the following shall apply:

$$\text{Error rate} \cdot \text{Excess percentage} = 0,3 \text{ ppm} \quad (\text{B.4})$$

This results in the permissible excess percentage of

$$\text{permissible excess percentage} = 0,3 \text{ ppm} / \text{error rate} \quad (\text{B.5})$$

The corresponding value for u_{1-p} is then determined for the excess percentage in the tabulated normal distribution.

For example, an error rate of 250 ppm results in a permissible excess percentage for the IP-system of 1 200 ppm. In a tabulated normal distribution, a value $u = 3,04$ is shown here.

Then the limit value for c_{ck} is as follows:

$$c_{ck \text{ grenz}} = \frac{u_{1-p}}{3} \quad (\text{B.6})$$

In the example with a 250 ppm error rate in the production process, this means a requirement of $c_{ck} \geq 1,01$.